

11-matching-flow: 匹配与网络流

(Matching and Network Flow)

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3 Theorems + 1 Algorithm

To **maximize** the size of a mathematical structure $\mathcal{S}$ in G



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Theorem

$\mathcal{S}$ is maximum *iff* G does not contain $\mathcal{S}$ -augmenting objects.

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Algorithm

Repeatedly finding $\mathcal{S}$ -augmenting objects until no more ones exist.

To **maximize** the size of a mathematical structure $\mathcal{S}$ in G



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To **minimize** the size of its **dual** mathematical structure $\mathcal{S}'$ in G

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Theorem (Weak Duality Theorem)

The size of a maximum $\mathcal{S} \leq$ The size of a minimum $\mathcal{S}'$

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Theorem (Strong Duality Theorem)

The size of a maximum $\mathcal{S} =$ The size of a minimum $\mathcal{S}'$

let's get
married
today



The Marriage Problem (Philip Hall, 1935)

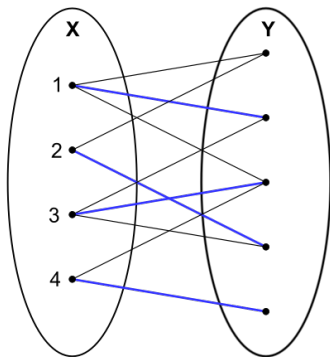
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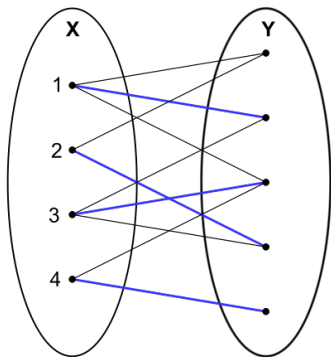
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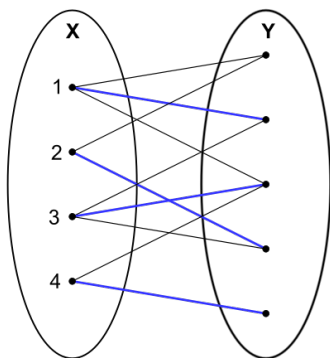
Philip Hall (1904 ~ 1982)





Definition (Matching (匹配))

A **matching** in a graph G is a set of edges with **no shared endpoints**.

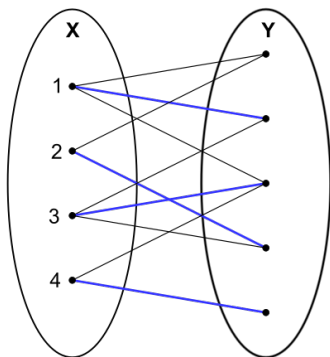


Definition (X -Perfect Matching (X -Saturating Matching))

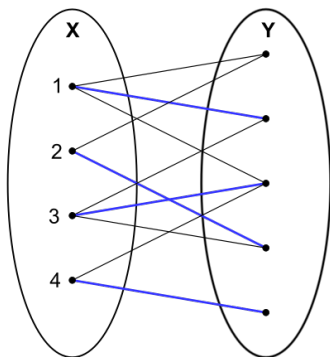
Let $G = (X, Y, E)$ be a bipartite graph.

An X -perfect matching of G is a matching which covers each vertex in X .

$$|X| \leq |Y|$$



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$$\forall W \subseteq X. |W| \leq |N(W)|$$

Theorem (Hall Theorem; 1935)

Let $G = (X, Y, E)$ be a bipartite graph. There is an X -perfect matching of G iff

$$\forall W \subseteq X. |W| \leq |N_G(W)|$$

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Theorem (Hall Theorem; 1935)

Let $G = (X, Y, E)$ be a bipartite graph. There is a X -perfect matching of G *iff*

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Definition (M -augmenting Paths)

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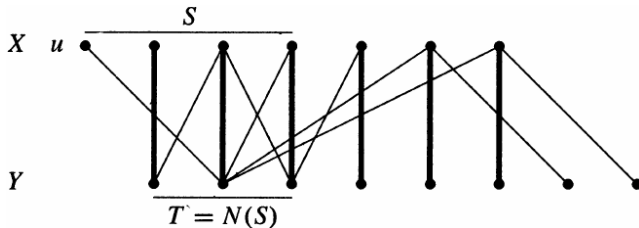
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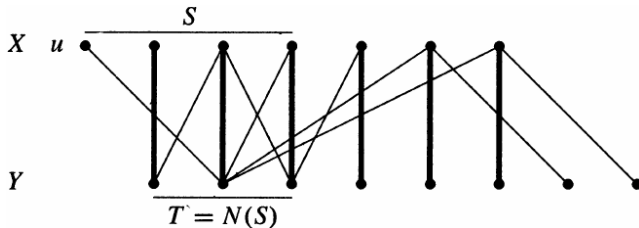
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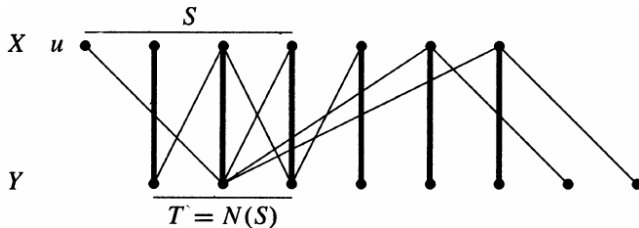


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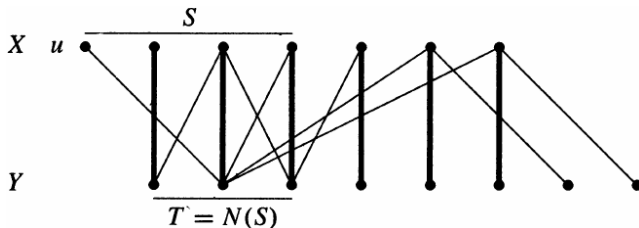
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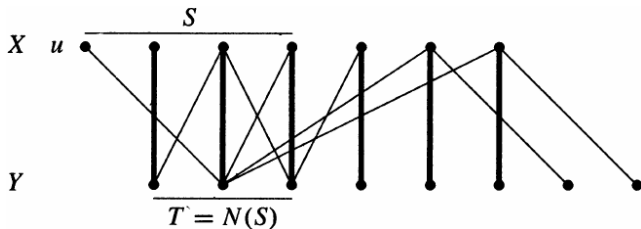
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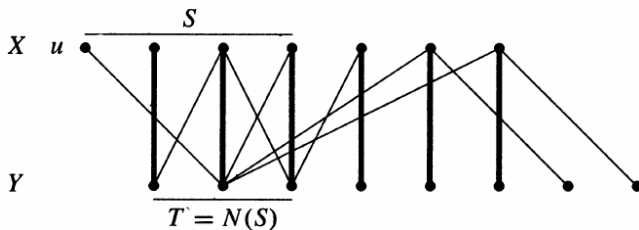
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Consider all *M -alternating paths* starting from u .

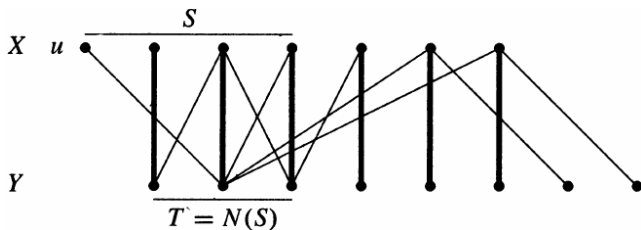


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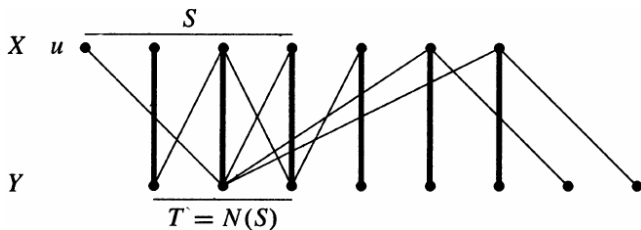


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$$T = N(S) \wedge |T| = |S - \{u\}|$$



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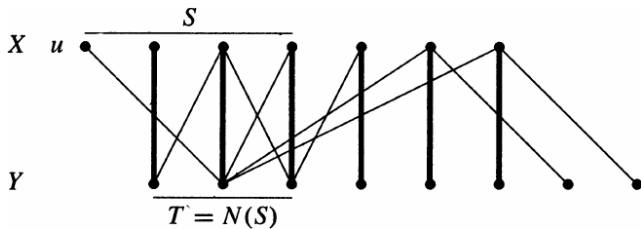
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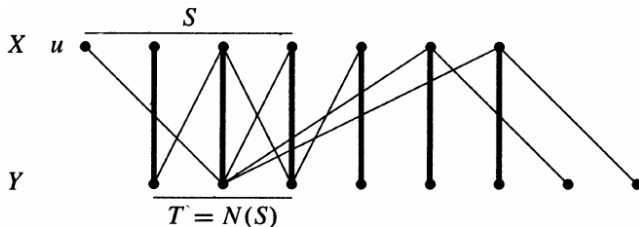
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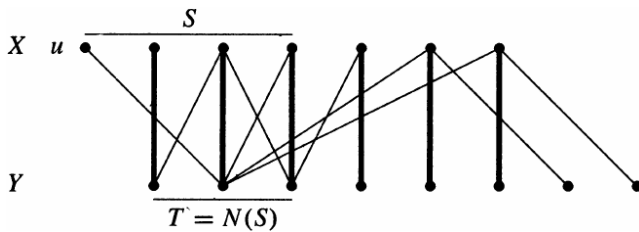
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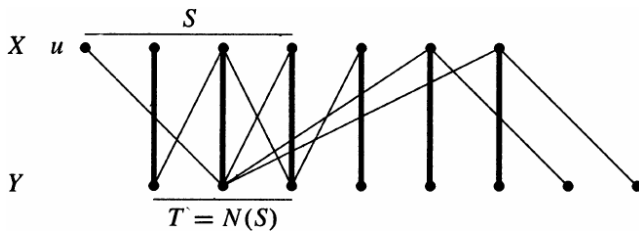
We show that there is a bijection from T to $S - \{u\}$.

M matches T with $S - \{u\}$.

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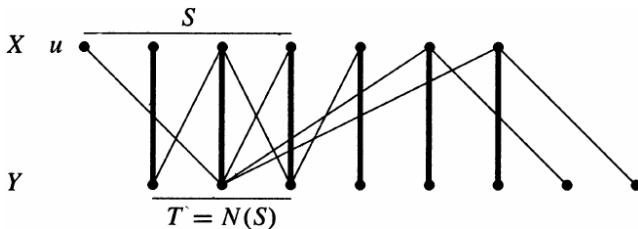


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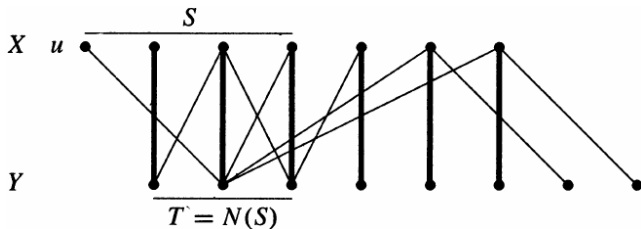
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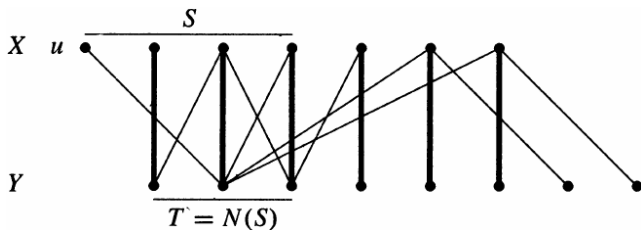


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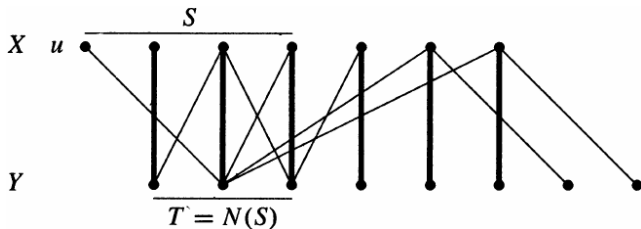


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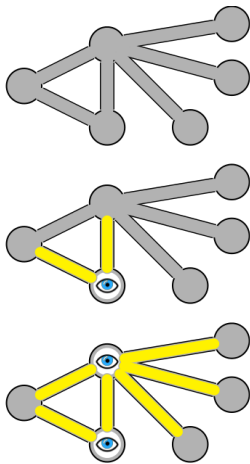
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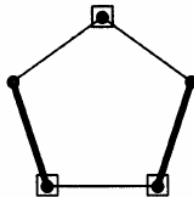
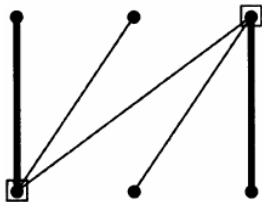
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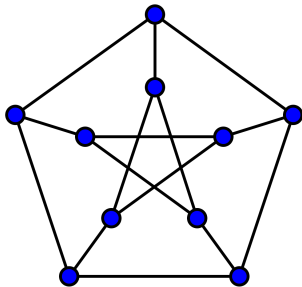
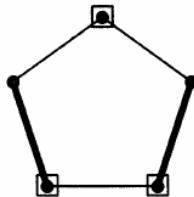
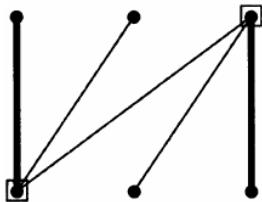
Consider the neighbors of $x \in S$: $x = u$, $x \in S - \{u\}$

Definitions (Vertex Cover (点覆盖))

A **vertex cover** of a graph G is a set $Q \subseteq V(G)$ that **covers** all edges.





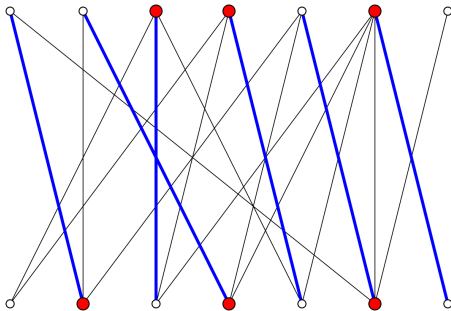


Theorem (Weak Duality Theorem (弱对偶定理))

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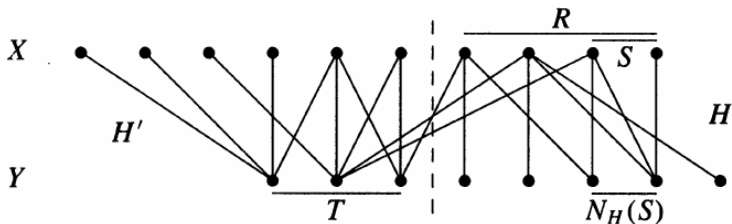


Theorem (König (1931), Egerváry (1931))

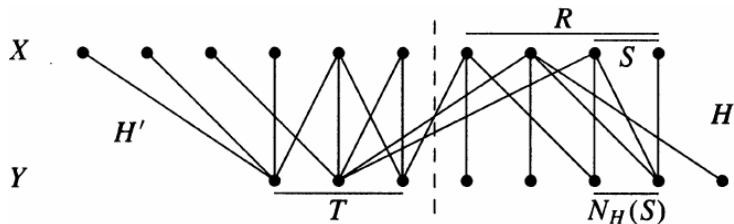
Let G be a *bipartite* graph. The maximum size of a matching in G
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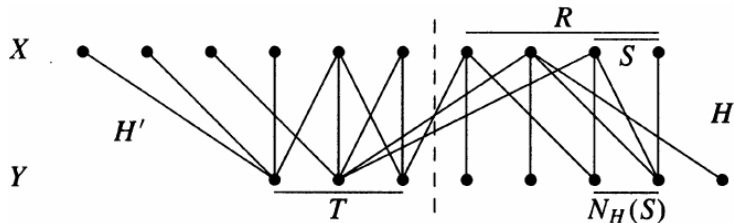


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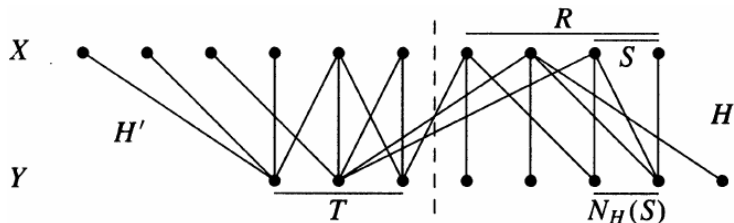


$$R = Q \cap X \quad T = Q \cap Y$$

$H \triangleq (R \cup (Y - T))$ -induced subgraph of G

$H' \triangleq (T \cup (X - R))$ -induced subgraph of G

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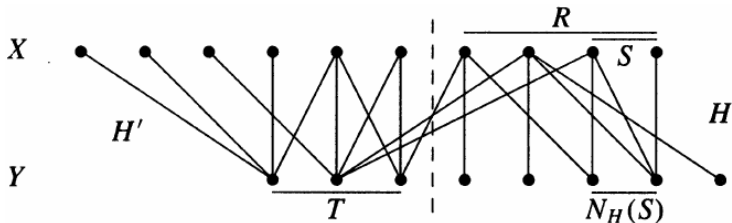
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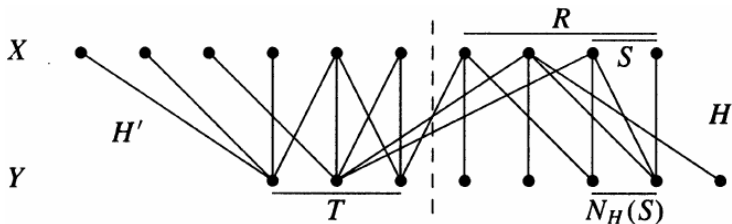
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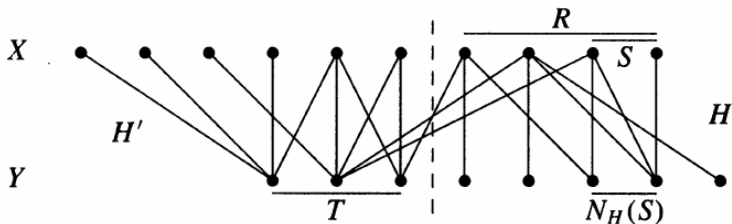
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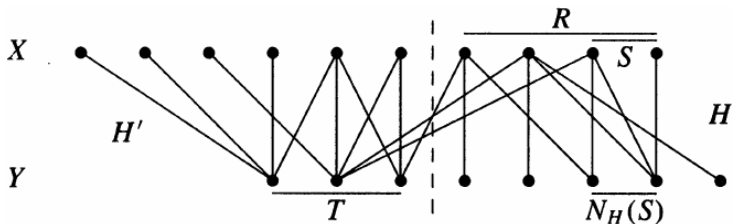


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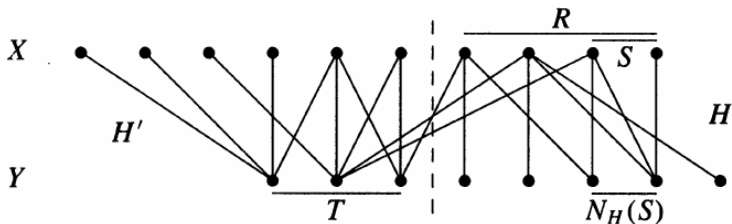
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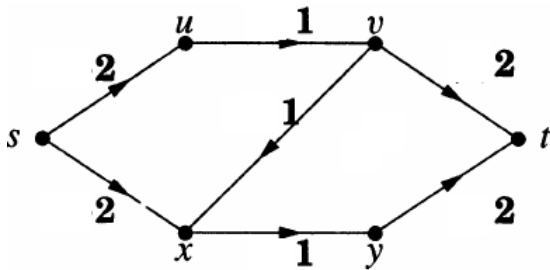
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$T \cup (R - S + N_H(S))$ is a smaller vertex cover than Q

Definition (Network (网络))

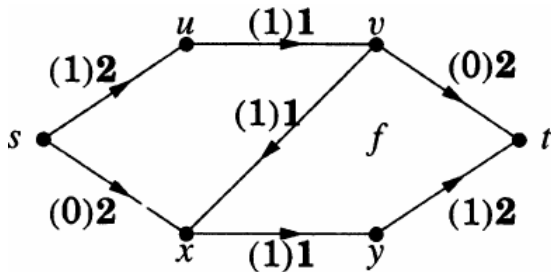
A **network** is a **digraph** with

- ▶ a distinguished **source vertex** s ,
- ▶ a distinguished **sink vertex** t ,
- ▶ a **capacity** $c(e) \geq 0$ on each edge e



Definition (Flow (流))

A **flow** f is a **function** that assigns a value $f(e)$ to each edge e .



Definition (Feasible Flow (可行流))

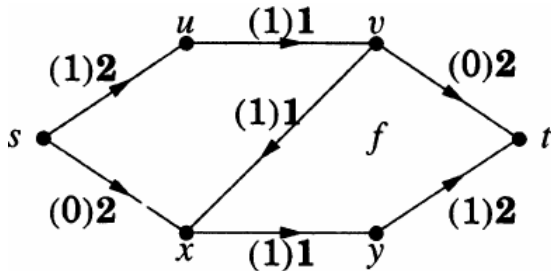
A flow f is **feasible** if it satisfies

Capacity Constraints:

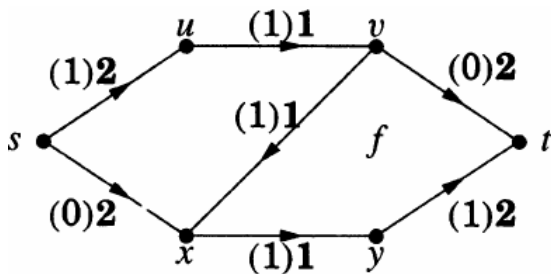
$$\forall e \in E. 0 \leq f(e) \leq c(e)$$

Flow Conservation:

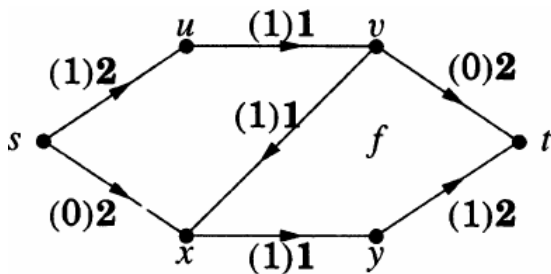
$$\forall v \in V. f^+(v) = f^-(v)$$



$$f^+(v) = \sum_{(v,w) \in E} f(v,w) \quad f^-(v) = \sum_{(u,v) \in E} f(u,v)$$

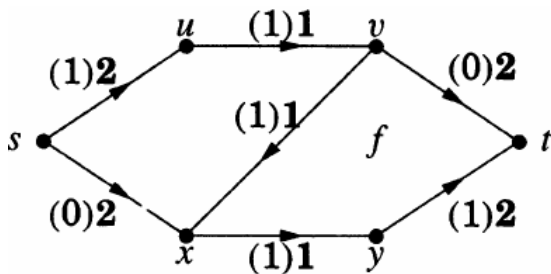


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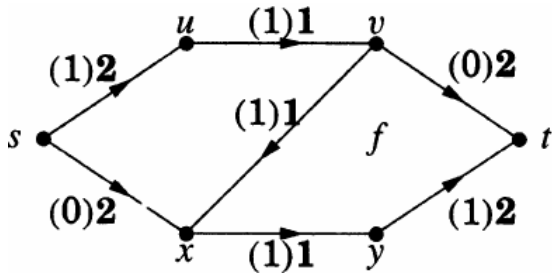


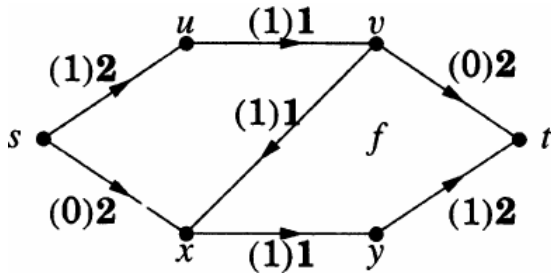
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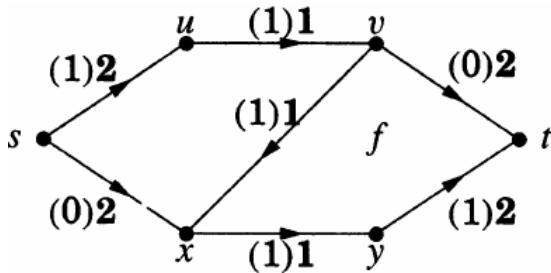


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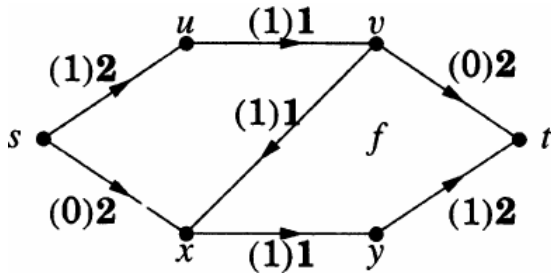


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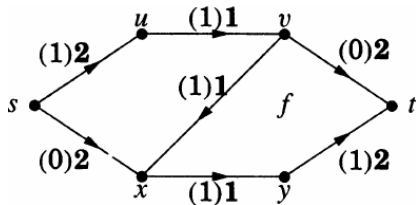
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Definition (Value (值))

The **value** $\text{val}(f)$ of a **flow** f is

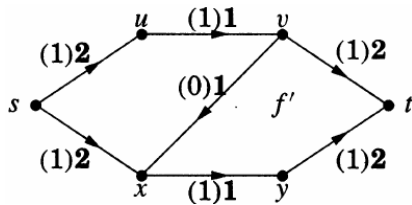
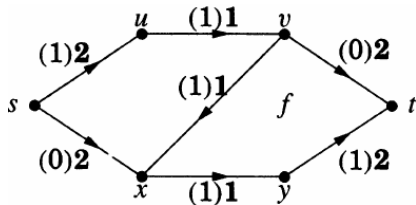
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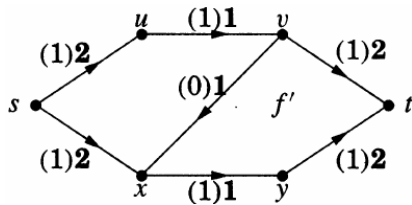
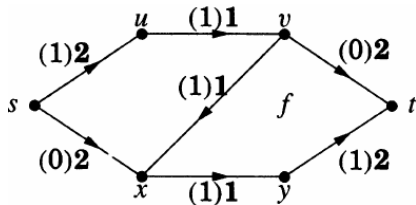
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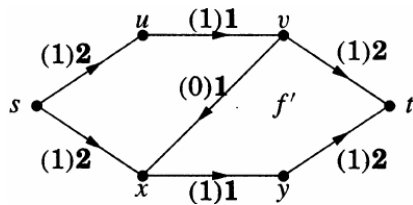
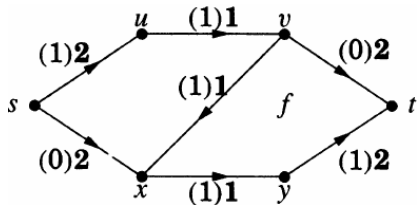
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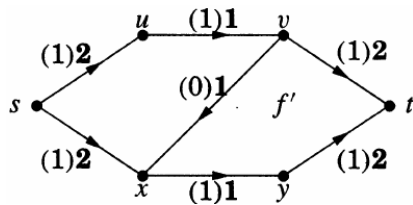
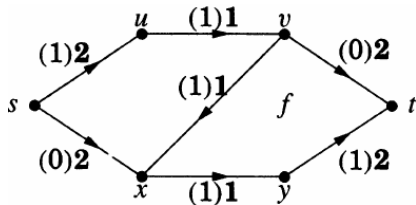
Definition (Maximum Flow (最大流))

A **maximum flow** is a **feasible flow** of maximum **value**.

$$s - x - v - t$$



$$s - x - v - t$$



Definition (f -augmenting Paths (增广路径))

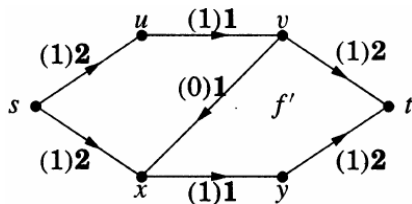
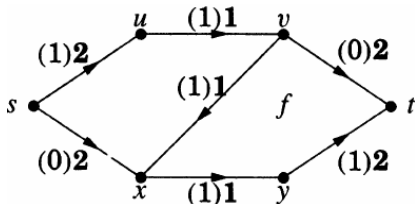
When f is a feasible flow, an **f -augmenting path** is a $s \sim t$ path P in the underlying graph such that for each edge $e \in E(P)$,

- (a) if P follows e in the forward direction, then $f(e) < c(e)$;
- (b) if P follows e in the backward direction, then $f(e) > 0$.

Definition (f -augmenting Paths)

Let P be an f -augmenting path.

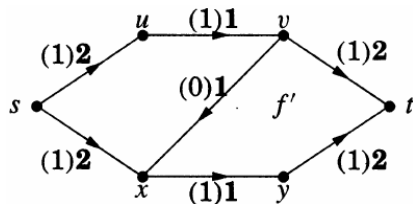
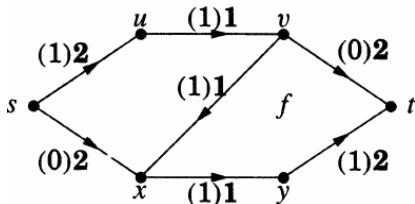
$$\epsilon(e) = \begin{cases} c(e) - f(e) & \text{if } e \text{ is forward on } P \\ f(e) & \text{if } e \text{ is backward on } P \end{cases}$$



Definition (f -augmenting Paths)

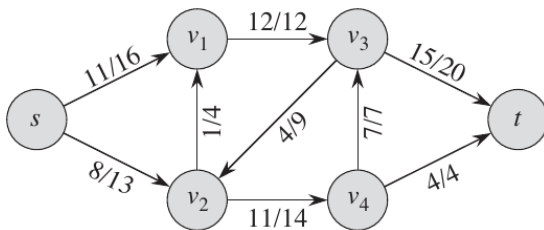
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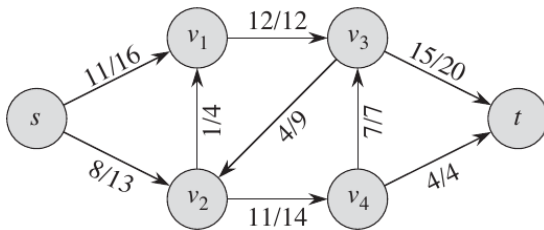
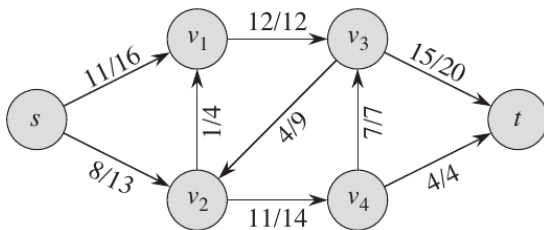
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An f -augmenting path leads to a flow with **larger** value.

$$\min_{e \in E(P)} \epsilon(e)$$

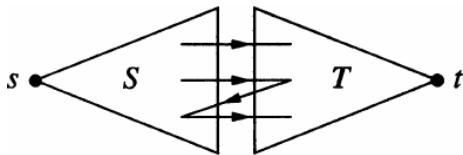


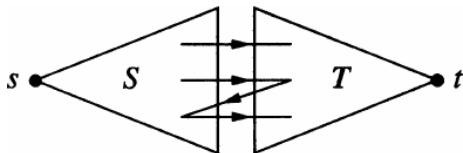


Definition (Source/Sink Cut (割))

In a network, a **source/sink cut** $[S, T]$ consists of the edges **from** a **source set** S **to** a **sink set** T , where

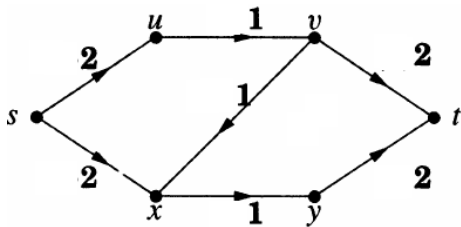
$$(T = V - S) \wedge (s \in S) \wedge (t \in T)$$

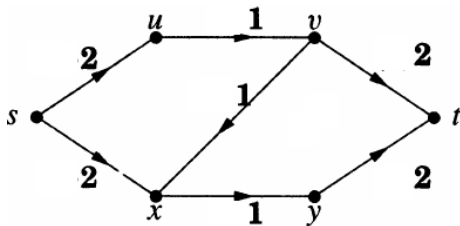




Definition (Capacity of Cut (割的容量))

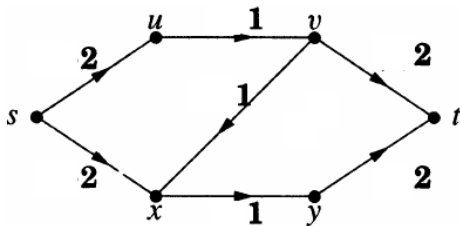
$$\text{cap}(S, T) = \sum_{u \in S, v \in T, uv \in E} c(u, v)$$





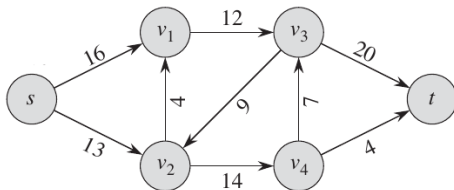
Definition (Minimum Cut (最小割))

A **minimum cut** is a **cut** of minimum value.



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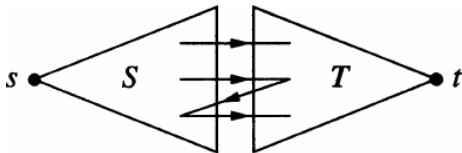
A **minimum cut** is a **cut** of minimum value.



Theorem (Weak Duality (弱对偶定理))

Let f be any feasible *flow* and $[S, T]$ be any source/sink *cut*.

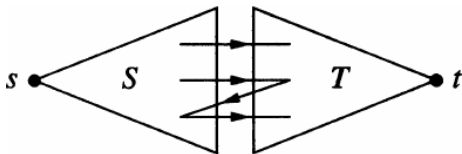
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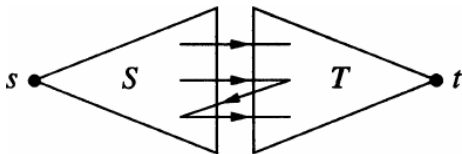


$$\text{val}(f) = f^+(S) - f^-(S)$$

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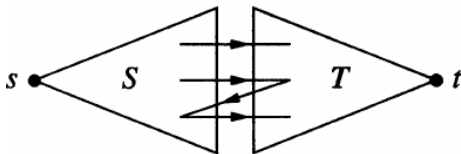


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$$\max_f \text{val}(f) \leq \min_{[S,T]} \text{cap}(S, T)$$

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f is maximum and $[S, T]$ is minimum

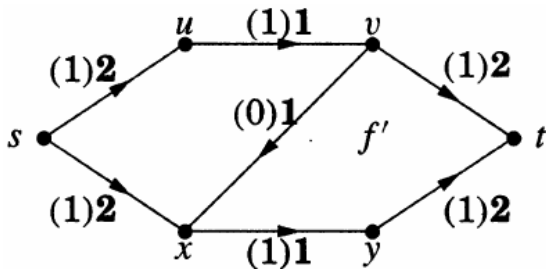
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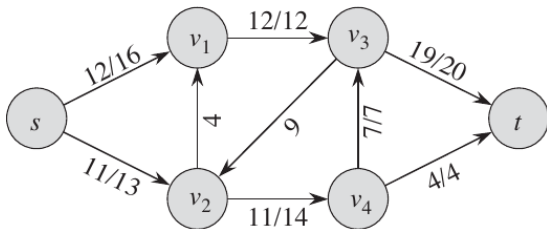
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Theorem (Max-flow Min-cut Theorem (Ford and Fulkerson; 1956))

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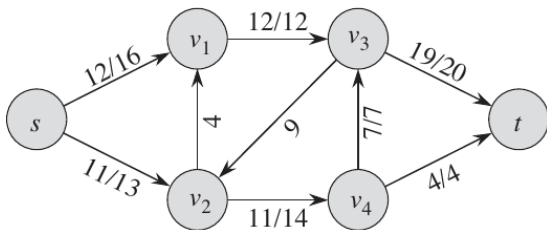
(Strong Duality)

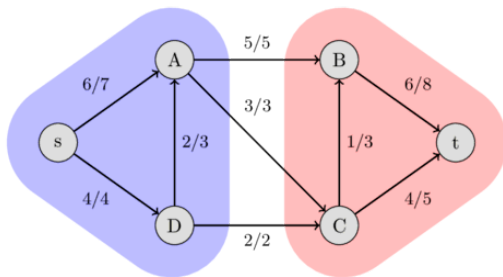
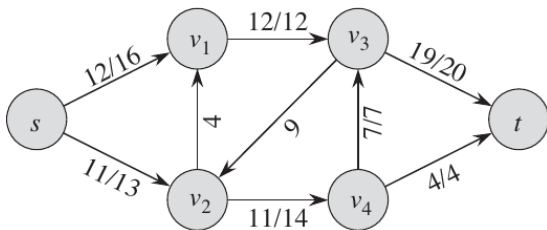


L. R. Ford Jr. (1927 ~ 2017)



D. R. Fulkerson (1924 ~ 1976)





Theorem

A feasible flow f is maximum iff there are no f -augmenting paths.

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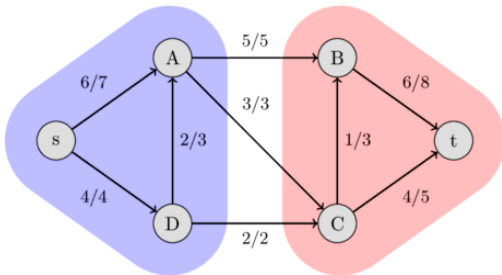
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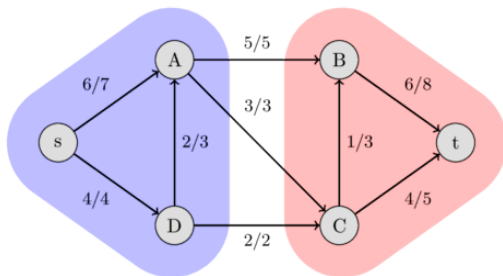
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$S \triangleq \{\text{the vertices reachable from } s \text{ along } \text{partial } f\text{-augmenting paths}\}$

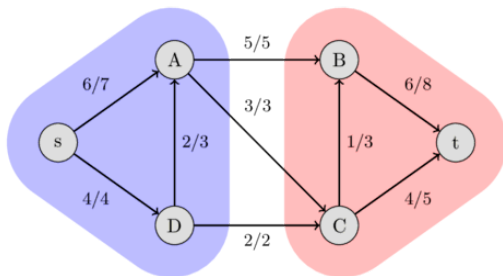
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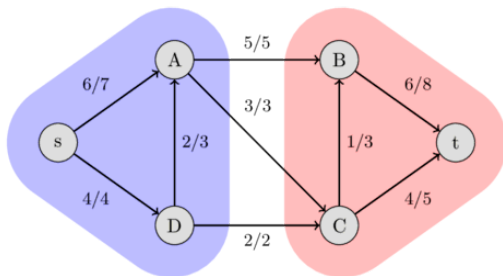
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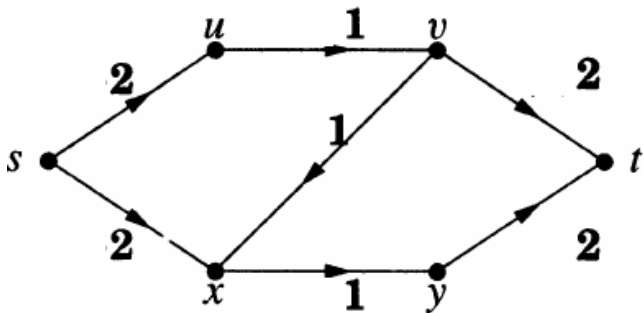
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The Ford-Fulkerson Method

Repeatedly finding f -augmenting paths until no more ones exist.

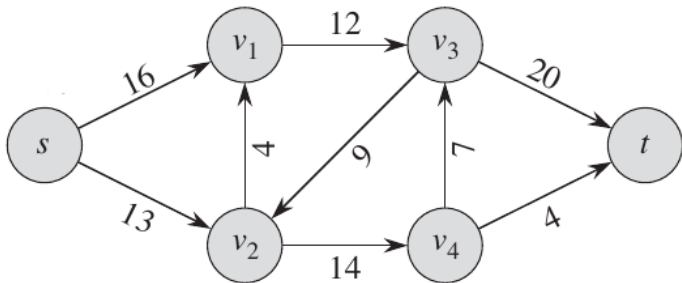
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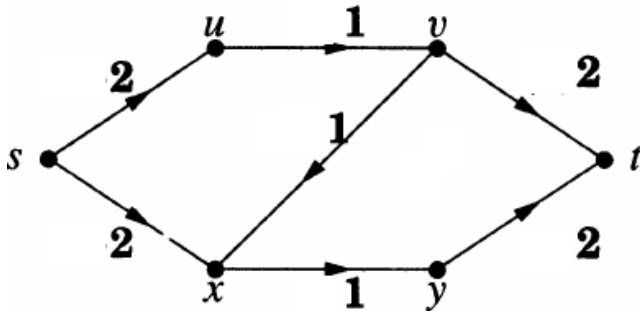


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Using **BFS** (Breadth-first Search) to find f -augmenting paths.

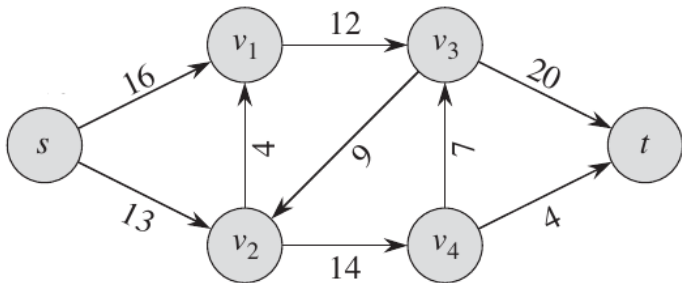
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Theorem (Hall Theorem; 1935)

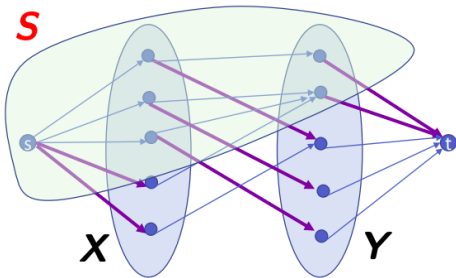
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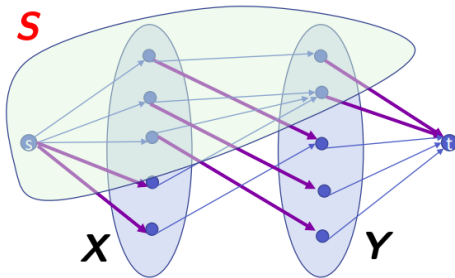


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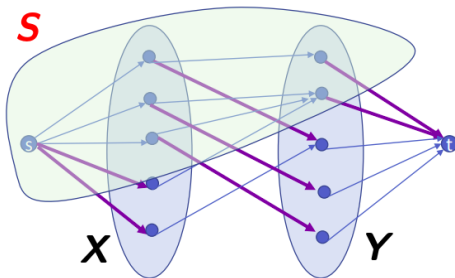
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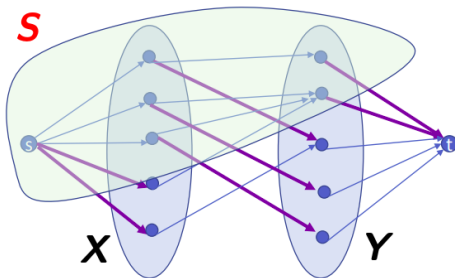
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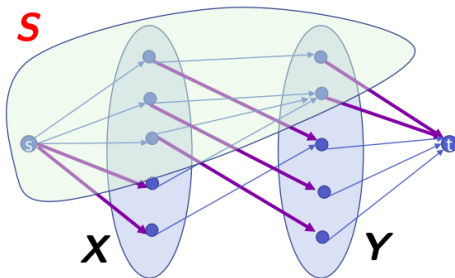


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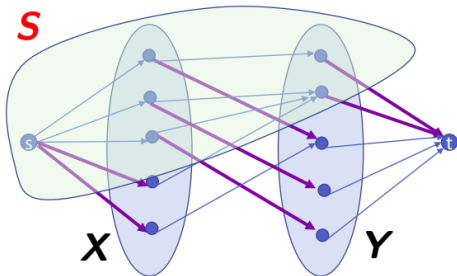
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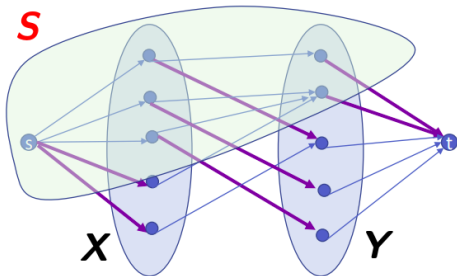
Therefore, we need to show that $\min_{[S, \bar{S}]} \text{cap}(S, \bar{S}) \geq |X|$.

Let $[S, \bar{S}]$ be a cut. We need to show that $\text{cap}(S, \bar{S}) \geq |X|$.



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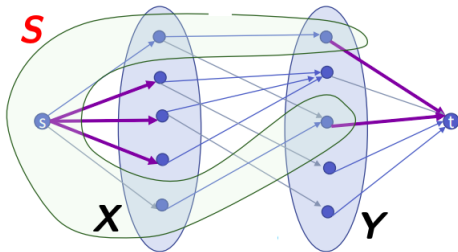
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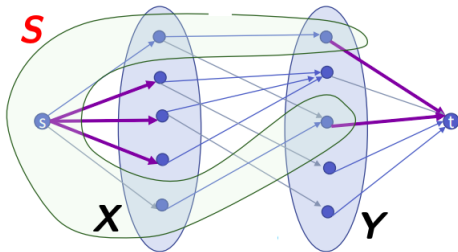
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$$\text{cap}(S, \bar{S}) = \infty \geq |X|$$

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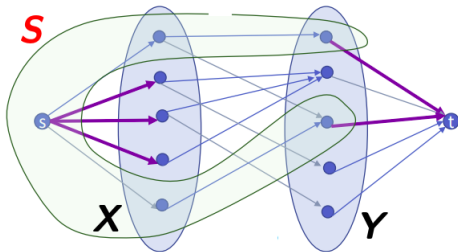


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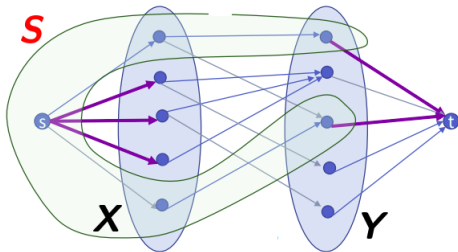
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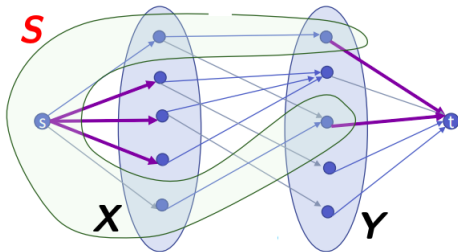
$$\begin{aligned} \text{cap}(S, \bar{S}) &= \sum_{u \in S, v \in \bar{S}} c(u, v) \\ &= \sum_{v \in \bar{S} \cap X} c(s, v) + \sum_{u \in S \cap Y} c(u, t) \end{aligned}$$

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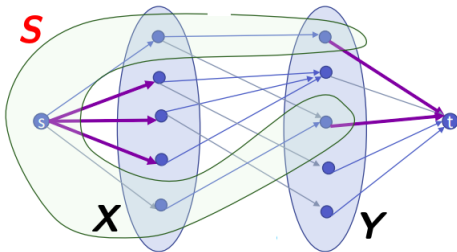
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$$\begin{aligned}
 \text{cap}(S, \bar{S}) &= \sum_{u \in S, v \in \bar{S}} c(u, v) \\
 &= \sum_{v \in \bar{S} \cap X} c(s, v) + \sum_{u \in S \cap Y} c(u, t) \\
 &= |X| - |S \cap X| + |S \cap Y| \\
 &\geq |X| - |S \cap X| + |N(S \cap X)|
 \end{aligned}$$

$$N(S \cap X) \subseteq (S \cap Y)$$



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 &= |X| - |S \cap X| + |S \cap Y| \\
 &\geq |X| - |S \cap X| + |N(S \cap X)| \geq |X|
 \end{aligned}$$

Theorem (König (1931), Egerváry (1931))

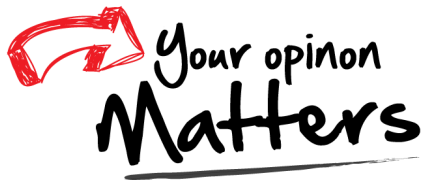
If G is a bipartite graph, then the maximum size of a matching in G equals the minimum size of a vertex cover of G

Theorem (König (1931), Egerváry (1931))

If G is a bipartite graph, then the maximum size of a matching in G equals the minimum size of a vertex cover of G



Thank
You!



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